CHEST DISEASES HOSPITAL | LUNG TRANSPLANT UNIT

Lung Transplantation Perioperative Protocol

Comprehensive Pathway: Admission to Discharge — ISHLT / IHLSC Aligned

Version 1.0
April 2026
Ref: CDH-LTU-PERIOP-02

Scope	Perioperative management of adult lung transplant recipients from admission through hospital discharge
Patient Population	Adults (≥18 years) undergoing single or bilateral lung transplantation at CDH
Related Protocols	CDH-LTU-ANAES-01 (Anaesthesia Protocol) • CDH-LTU-PPO-03 (Post-operative Orders) • CDH-LTU-IMPL-01 (LTU Implementation Protocol)

CROSS-REFERENCE This protocol is to be read alongside the CDH LTU Anaesthesia Protocol (CDH-LTU-ANAES-01) for intraoperative anaesthesia management and the Post-operative Orders (CDH-LTU-PPO-03) for ICU order sets. Clinical judgment remains paramount in the presence of anatomical variations or complications.

PHASE 1 — Day 0 — Pre-Operative: ADMISSION & PRE-OPERATIVE PREPARATION

1.1 The Call & Admission

Patient is admitted to the pre-transplant/thoracic isolation unit upon organ offer acceptance. All admissions must be treated as emergent.

Task	Detail
Immediate Investigations	STAT CBC + differential, comprehensive metabolic panel, coagulation profile (PT/INR/PTT), viral serologies, Type & Crossmatch (PRBCs, FFP, platelets prepared).
Diagnostics	STAT 12-lead ECG and upright CXR to establish baseline and rule out acute infection.
NPO Status	Strict NPO enforced immediately on admission.
Physical Examination	Full physical examination focusing on respiratory status; document any active infection signs.
Informed Consent	Consent obtained for transplant procedure and possible ECMO/CPB use.
Anticoagulants	Withhold standard anticoagulants.
Induction Immunosuppression	Prepare induction immunosuppression (Basiliximab/Alemtuzumab) per protocol to be given prior to or during procedure.
Transplant Coordinator	Notify transplant coordinator; confirm cold ischaemia clock. Relay OR team readiness.

PHASE 2 — Day 0 — Operating Room: INTRAOPERATIVE STANDARDS & SURGICAL STEPS

SEE ALSO For complete anaesthesia management including monitoring, ECMO protocols, and emergency procedures, refer to CDH-LTU-ANAES-01 (Anaesthesia Protocol). This section covers surgical standards and implantation technique.

2.1 Pre-Operative Anaesthesia & ECLS Readiness

- Left-sided double-lumen endotracheal tube (DLT) with immediate bronchoscopic confirmation of position.
- Arterial line, CVC, pulmonary artery catheter with continuous SvO_2, and continuous TEE.
- ECLS Readiness: Central V-A ECMO preferred over CPB. Femoral vessels prepped and exposed routinely before incision if peripheral cannulation is anticipated.
- Confirm cold ischaemia time with transplant coordinator before skin incision.

2.2 Patient Positioning & Incision

Procedure	Position & Incision
Bilateral Sequential Lung Transplant (BSLTx)	Supine with arms extended and elevated. Bilateral anterolateral thoracotomy (clamshell) through 4th or 5th intercostal space with transverse sternotomy.
Single Lung Transplant (SLTx)	Lateral decubitus. Standard posterolateral thoracotomy.

2.3 Surgical Implantation Sequence — Posterior-to-Anterior

SURGICAL STANDARD The implantation sequence strictly follows posterior-to-anterior anatomical orientation: Bronchus → Pulmonary Artery → Pulmonary Veins (Atrial Cuff). Meticulous haemostasis is critical throughout.

Step	Procedure	Technical Details
1	**Native Pneumonectomy**	• Isolate and divide pulmonary artery and veins using vascular staplers. • Divide mainstem bronchus proximal to the upper lobe take-off. • Remove native lung; thoroughly irrigate pleural cavity with warm antibiotic solution. • Send pleural fluid for immediate microbiological culture.
2	**Bronchial Anastomosis**	• Trim donor and recipient bronchi closely to secondary carinae to minimise ischaemic watershed areas. • End-to-end anastomosis with 4-0 absorbable monofilament (PDS): – Membranous portion: continuous running suture. – Cartilaginous portion: interrupted figure-of-eight or continuous running with precise mucosal approximation. • Wrap anastomosis with peribronchial tissue or vascularised pedicle (intercostal muscle flap, thymic tissue) if ischaemic risk is high.
3	**PA Anastomosis**	• Ensure appropriate donor PA length to avoid kinking post-reperfusion. • End-to-end anastomosis: 5-0 continuous non-absorbable monofilament (Prolene). • Leave anterior aspect of suture line loose ("growth factor") to prevent stricture upon reperfusion and distension. • Flush PA gently prior to tying final knot to clear intraluminal thrombus or air.
4	**LA Cuff Anastomosis**	• Apply Satinsky clamp to recipient left atrium, encompassing pulmonary vein stumps. • Open recipient pulmonary veins to create unified atrial cuff. • Anastomose donor LA cuff to recipient LA cuff: continuous 4-0 Prolene suture.
5	**De-airing & Reperfusion**	• **CRITICAL: Reperfusion must be performed GRADUALLY over 10–15 minutes. Controlled slow reperfusion minimises endothelial shear stress and limits PGD.** • Before fully tying LA anastomosis: briefly release PA clamp to flush blood and air via LA suture line (retrograde de-airing). • Tie LA anastomosis securely. • Partially release PA clamp. Wait for haemodynamic stabilisation. Administer Methylprednisolone 500 mg IV prior to full reperfusion. • Assess implanted lung: direct vision for expansion, compliance, and homogeneous colour.

PHASE 3 Days 0–3 — Cardiothoracic ICU — IMMEDIATE POST-OPERATIVE ICU MANAGEMENT

SEE ALSO: CDH-LTU-PPO-03 For complete ICU order set, refer to CDH-LTU-PPO-03 (Post-operative Orders). All patients must be admitted to Cardiothoracic ICU in STRICT ISOLATION.

System	Protocol Guidelines
Respiratory — Ventilation	Protective lung ventilation: Vt 6 mL/kg DONOR ideal body weight. PEEP 5–8 cmH_2O. Target extubation within 12–24 hours to prevent VAP and reduce positive pressure effects on bronchial healing.
Haemodynamics	Strict "dry" state. Target CVP <8 mmHg. MAP 65–80 mmHg via vasopressors (norepinephrine preferred over fluid boluses). Use loop diuretics/CRRT aggressively for fluid overload.
Immunosuppression	Triple therapy initiation: CNI (Tacrolimus target 10–15 ng/mL months 1–3) + Antimetabolite (MMF 1000 mg BID) + tapering Corticosteroids. Check tacrolimus trough daily at 06:00.
Infection Prophylaxis	Bacterial: broad-spectrum (Piperacillin/Tazobactam) tailored to donor/recipient cultures. Fungal: inhaled Amphotericin B or systemic antifungal. Viral: Valganciclovir (CMV prophylaxis per serostatus). PCP: Trimethoprim/Sulfamethoxazole.
Chest Tubes	Maintain on −20 cmH_2O suction initially. Monitor for haemorrhage (notify surgeon if >150 mL/hr × 2 hours). Transition to water seal when drainage serous and <200 mL/24h with no air leak.
Analgesia	Thoracic epidural preferred. If epidural contraindicated (anticoagulation): bilateral ESPB. Paracetamol 1 g IV q6h scheduled. Fentanyl PCA if no epidural. NSAIDs CONTRAINDICATED.
Fluid Management	D5W at 30 mL/hr to KVO. Strict I/O monitoring. Target negative fluid balance. Furosemide IV PRN to maintain CVP <8 mmHg.
Renal	Hourly urine output monitoring. Target ≥0.5 mL/kg/hr. Nephrology consult early if oliguria or eGFR decline.

3.1 Bronchoscopy Schedule — Post-operative

Timing	Type	Purpose
Day 0/1 — Post-op	Bedside (routine)	Inspect anastomoses, clear secretions, confirm airway patency.
Day 1–3 (if concern)	Therapeutic / diagnostic	Atelectasis, mucus plugging, infiltrate assessment, culture.
1 Month (scheduled)	TBBx + BAL	Protocol surveillance; rejection grading; culture. See Year-1 Form (CDH-LTU-YEAR1-03).

3.2 Primary Graft Dysfunction (PGD) — ICU Grading & Management

ISHLT PGD DEFINITION PGD is assessed at 0, 24, 48, and 72 hours post-reperfusion. Diagnosis: bilateral radiographic infiltrates + PaO_2/FiO_2 decline within 72 hours, excluding other causes (anastomotic complication, haemothorax, lobar torsion, cardiogenic oedema). Grade represents worst documented score.

Grade	PaO_2/FiO_2	Management
0	≥300 mmHg	Standard lung-protective ventilation. Routine monitoring. Early extubation protocol as planned.
1	200–299 mmHg	Continue protective ventilation. Review fluid balance. Increase monitoring frequency (ABG q2h). Diuresis.
2	100–199 mmHg	Increase PEEP (8–12 cmH_2O); Pplat <30 cmH_2O; prone positioning if refractory. Consider ECMO consultation.

3	<100 mmHg	VV-ECMO or VA-ECMO per institutional protocol. TV 4–6 mL/kg IBW; PEEP 8–12 cmH_2O; Pplat <30 cmH_2O. Minimise FiO_2. Inhaled nitric oxide (iNO) may be added.

3.3 CMV Prophylaxis — Serostatus-Guided

Donor (D)	Recipient (R)	Protocol
D+	**R−**	Highest risk. Valganciclovir 900 mg PO daily (or IV Ganciclovir 5 mg/kg per renal function) for minimum 6 months. CMV PCR weekly ×4 then monthly.
D+	**R+**	High risk. Valganciclovir 900 mg PO daily for minimum 3 months. CMV PCR monitoring monthly.
D−	**R+**	Intermediate risk. Valganciclovir 900 mg PO daily for minimum 3 months.
D−	**R−**	Low risk. Acyclovir prophylaxis. CMV PCR PRN if symptomatic.

PHASE 4 — STEP-DOWN WARD & REHABILITATION
Days 4–14

4.1 Chest Tube Management

- Maintain on −20 cmH_2O suction initially.
- Transition to water seal when drainage is serous and <200 mL/24h with no air leak.
- Remove chest tubes sequentially: right pleural → left pleural → mediastinal (if applicable).
- Daily CXR for assessment of lung expansion and fluid collections.

4.2 Airway Clearance Programme

- Aggressive bronchopulmonary hygiene from Day 1 of extubation.
- Incentive spirometry every hour while awake.
- Nebulised bronchodilators and mucolytics as indicated.
- Chest physiotherapy: percussion and vibration q4h while intubated.
- Therapeutic bronchoscopy if atelectasis persists beyond 24–48 hours.

4.3 Early Mobilisation Programme

Stage	Activity Target
Stage 1 (Day 1)	Out of bed to chair (if haemodynamically stable). Passive limb exercises.
Stage 2 (Days 2–3)	Standing with assistance. Ambulation with physiotherapist to bedside.
Stage 3 (Days 4–7)	Hallway ambulation with pulse oximetry monitoring. Progressive increase in distance.
Stage 4 (Days 8–14)	Stationary cycling. Stair climbing. Independent ambulation with continuous SpO_2 monitoring.

⚠ **SAFETY NOTE** Continuous pulse oximetry during ALL exertion phases. Halt activity if SpO_2 <90% or patient reports significant dyspnoea.

4.4 Nutritional Support

- Early enteral feeding: commence immediately upon extubation if swallow assessment is passed.
- If prolonged ventilation (>48h): insert nasogastric/Dobhoff tube; begin enteral feeds within 24–48 hours of admission.
- Dietitian consultation: ensure adequate caloric and protein intake for wound healing.
- Blood glucose monitoring: steroid-induced hyperglycaemia is common; insulin protocol as per ICU policy.
- Target protein: 1.5–2.0 g/kg/day. Caloric target: 25–30 kcal/kg/day.

4.5 Immunosuppression Monitoring — Step-Down Phase

Agent	Target / Dose	Monitoring
Tacrolimus (CNI)	Months 1–3: 10–15 ng/mL trough Months 4–12: 8–12 ng/mL trough	Daily 12h trough at 06:00
Mycophenolate (MMF)	1000 mg BID (IV or PO) Reduce if leucopenia or GI intolerance	CBC weekly. GI monitoring.
Prednisolone	Taper per institutional protocol from Methylprednisolone induction	Blood glucose monitoring

PHASE 5 Days 14–21+ DISCHARGE PLANNING & PATIENT EDUCATION

5.1 Mandatory Discharge Criteria

DISCHARGE GATE ALL criteria below must be met before discharge. No patient to be discharged without therapeutic tacrolimus trough level confirmed for at least 48 consecutive hours and confirmed outpatient follow-up within 7 days.

	Discharge Criterion
☐	Ambulatory independently or with baseline assistance.
☐	Acceptable oxygen saturation on room air (or at safe, stable supplemental O_2 baseline).
☐	Surgical incisions healing with no signs of active infection.
☐	Therapeutic tacrolimus trough levels achieved and documented for ≥48 consecutive hours.
☐	Stable spirometry (FEV_1 ≥50% of predicted or at acceptable post-operative baseline for underlying diagnosis).
☐	Afebrile for ≥48 hours without new antibiotics.
☐	Outpatient transplant clinic appointment confirmed within 7 days of discharge.
☐	Outpatient pulmonary rehabilitation programme booked.
☐	Bronchoscopy at 1 month scheduled.
☐	Patient and caregiver fully educated on all medications, monitoring, and warning symptoms.

5.2 Pre-Discharge Patient Education Checklist

✓	Education Topic	Key Points to Cover
☐	**Immunosuppression adherence**	Complete understanding of the triple therapy schedule. Emphasise: never stop medications without contact with transplant team.
☐	**Home monitoring protocol**	Daily temperature, blood pressure, weight, and home spirometry (FEV_1 and FVC). Reinforce action thresholds for each.
☐	**Warning signs**	Fever >38°C, SpO_2 decline, spirometry drop >10%, change in sputum, rash, neurological symptoms, oedema.
☐	**Infection prevention**	Hand hygiene, avoid crowded spaces, sun protection, pet restrictions, food safety (immunocompromised diet).
☐	**Surgical precautions**	Thoracotomy/sternal lifting restrictions, wound care, safe body mechanics for 6–8 weeks.
☐	**Sun protection & skin surveillance**	Post-transplant skin cancer risk. Annual dermatology review. Daily SPF 50+ and protective clothing.

☐	**Drug interactions**	Common OTC medications, herbal supplements, and foods (grapefruit) that interact with tacrolimus/cyclosporin.
☐	**Follow-up schedule**	Clinic visit within 7 days. Bronchoscopy at 1 month. Confirm next outpatient appointment before leaving.

SECTION 6: PROTOCOL FAMILY & CROSS-REFERENCES

Protocol	Reference Number	Scope
LTU Implementation Protocol	CDH-LTU-IMPL-01	Unit governance, mandate, clinical pathway overview, QAPI. Parent document.
Perioperative Protocol (this document)	CDH-LTU-PERIOP-02	Admission to discharge surgical and clinical pathway.
Anaesthesia Protocol	CDH-LTU-ANAES-01	Intraoperative anaesthesia, ECMO, monitoring, and emergency protocols.
Post-operative Orders (PPO)	CDH-LTU-PPO-03	ICU order set: ventilation, fluids, immunosuppression, prophylaxis, analgesia.
MDT Decision Form	CDH-LTU-MDT-05	Listing decisions, MDT minutes, decision documentation.
Year-1 Follow-Up Form	CDH-LTU-YEAR1-03	Post-transplant clinic surveillance form (months 1, 3, 6, 9, 12).
Annual Follow-Up Form	CDH-LTU-ANNUAL-04	Post year-1 annual surveillance with CLAD/DSA/dd-cfDNA.

SECTION 7: DOCUMENT CONTROL

Version	Details
1.0	Initial release — April 2026. Merged from v1 (surgical technique SOP) and v2 (admission-to-discharge pathway). Chest Diseases Hospital, LTU.
Review Due	April 2027 or following any critical incident or significant protocol deviation.
Approved By	Head of Lung Transplant Unit, CDH / Head of Cardiothoracic Surgery, CDH